(especially in the lower levels of the tree) are null references[9].

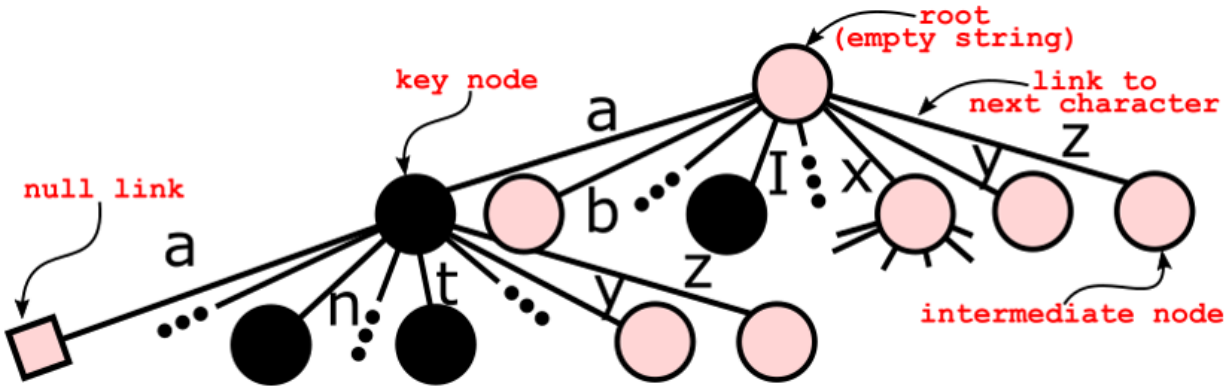


Figure 6.2 The structure of a Trie. Words are encoded in the tree one using edges, each edge corresponds to a single character, and each node n is associated with a single word, the one obtained by joining the characters associated with the edges in the path from the root to n: the root node corresponds to the empty string (because no edge is traversed), the leftmost leaf corresponds to "aa", etc… Not all the paths make meaningful words, and indeed not all the nodes store the word associated with them: only filled, black nodes (called "key nodes") mark words stored in the trie, while hollow nodes, aka "intermediate nodes", corresponds to prefixes of words stored in the trie. Notice that all leaves should be key nodes.

Nevertheless, this representation looks terrible on paper, to represent tries in diagrams: too many links, too many nodes, resulting in chaos.

That's why we will use an alternative representation, shown in figure 6.3: we'll only show links to actual trie nodes,